From Interpretable to Black-Box: Dynamic Model Identification for Small USV Control Design

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***Abstract*—Effective low-level feedback control of small underactuated uncrewed surface vehicles (USVs), specifically vessels under seven meters with a single thruster and rudder, requires dynamic models that provide both actionable design insight and predictive accuracy. Tow-tank testing is impractical at this scale; parameters must be identified solely from limited field trials. Existing system identification methods span a wide range of model complexity but offer little guidance on how much fidelity is enough, because sufficiency is only meaningful relative to a stated purpose. Absent that specification, the default is a bias toward more fidelity, since the benefit of an added term is readily quantified while its costs in identifiability, experimental burden and obscured design insight are not.**

We propose a methodology built around a deliberate spectrum of three model levels, evaluated on common experimental data: (1) first-order, linear, single-input single-output (SISO) transfer functions, Nomoto-style models for surge and yaw that are immediately interpretable and support controller architecture intuition, but risk oversimplification when considered in isolation; (2) a physics-based nonlinear state-space formulation incorporating quadratic hydrodynamic drag; and (3) a Nonlinear AutoRegressive with eXogenous inputs (NARX) multilayer perceptron. Rather than seeking a single best model, the spectrum is used collectively: simple models anchor human intuition; structured nonlinear models expose which terms are dynamically salient; and the black-box model bounds the benefit of added complexity.

Preliminary results from field experiments demonstrate the approach. Levels 1 and 2 are identified from a single step-response trial; Level 3 is trained on a larger corpus of manual-mode records and evaluated on that same trial. First-order SISO models achieve $R^2 = 0.93$ (surge, $\tau = 1.95$ s) and $R^2 = 0.83$ (yaw rate, $\tau = 0.47$ s), providing direct controller tuning parameters. Cross-model diagnostics identify quadratic drag as the dominant surge nonlinearity and reveal that speed-dependent rudder effectiveness, a key coupling for underactuated control design, is not resolvable from a single short trial, motivating targeted data collection. A margin-based diagnostic makes the sufficiency question computable, reporting the speed range over which the first-order model is sufficient for the control design. The NARX achieves one-step prediction $R^2 = 0.998$ for speed and $R^2 = 0.989$ for yaw rate on the common trial; recursive simulation on that trial falls to $R^2 = -1.988$ and $R^2 = 0.259$. The cause is measurable rather than generic: the network never acquired a steady-state throttle-to-speed relationship, settling below zero at a throttle where the vessel measurably does 1.22 m/s, because the records it was fitted to almost never dwell at equilibrium. The training runs never held throttle steady long enough for the vessel to settle, so the network was never shown which speed belongs to which throttle. A physics-based model does not have to be shown: setting the acceleration to zero in its equation of motion returns that relationship directly. One-step accuracy is a poor guide to simulation accuracy, and reporting it alone will not reveal this. High-speed segments are retained for subsequent regime-spanning analysis.

***Index Terms*—uncrewed surface vehicle, system identification, model spectrum, control-oriented modeling, robustness margin, nonlinear dynamics, underactuated vessels**

I. INTRODUCTION

Small uncrewed surface vehicles, here taken as vessels under seven meters driven by a single thruster and rudder, are increasingly deployed for survey, inspection and autonomy research. Every one of those missions requires a working low-level controller: a speed loop on the throttle and a yaw-rate loop on the rudder. In production autopilots such as ArduPilot these are proportional-integral loops with an added feedforward term. Designing, simulating and tuning these control solutions is the engineering task this paper addresses.

The system identification literature offers everything from two-parameter response models to maneuvering polynomials carrying dozens of hydrodynamic coefficients. What it does not offer is guidance on which of the models is best suited for controller design. Fixing the compensator structure makes that question answerable, because it turns concrete: does a change of model level move the gains, the achievable bandwidth or the tuning procedure of that loop? A model that alters none of the three criteria, no matter how much better the fit, does not change the design.

Using the most detailed, highest fidelity, model available is not a safe default. Added model terms degrade the conditioning of the fit [1], [2], and each new term needs excitation to identify it, which converts fidelity directly into field trials, and many model terms are not identifiable from the trials a small vessel can run. Sonnenburg and Woolsey [3] evaluate a cost function on held-out maneuvers, tabulate it per model and per regime, then make the final selection by judgment, discarding the best-scoring model because its two extra parameters are not worth the slight edge in turn-rate estimation. Skelton put the criterion in control terms in 1989, that model errors need not be small but only appropriate for the control design [4].

Our scope is the single thruster plus rudder vessel with field-identified parameters. Underactuated rudder steering is the interesting case, because control authority is an explicit function of forward speed, so the steering and speed channels cannot be treated as independent design problems.

We identify three model levels from common field data, compare them with cross-model diagnostics, and give a margin-based test that reports the speed range over which added fidelity stops changing the loop design. The deliverable is a procedure rather than a model: the diagnostics say which dynamics are present, and the margin test says which of them reach the compensator. Prior identification studies for this vehicle class compare candidate structures and then select one, after which the rest stop contributing; we keep the set and report what each one adds. The models buy loop structure, verification in simulation, relative comparison of candidate variants and a rehearsed tuning procedure. They do not buy gains that transfer without field tuning, and we say so plainly rather than let a reader infer otherwise.

II. Closely Related Work

The minimal control-oriented description of a steered vessel is Nomoto's first-order steering model [5], still the standard starting point in the modern formulation [6], with the identification tradition founded by Åström and Källström [7]. Physics-based maneuvering models such as Abkowitz's [8] include coefficients with physical meaning but resist identification from field data. Over the maneuvers a small vessel can realistically perform the regressors are nearly collinear, so many different coefficient sets fit the data about equally well and no single coefficient can be trusted on its own, however well the model predicts as a whole [1], [2], [9].

For small craft the closest prior work is Sonnenburg and Woolsey [3], [10], who identify and compare a family of models for a small planing craft and evaluate them for control design, with related work spanning autonomous surface craft and small workboats [11]–[15]. The common pattern is a comparison across candidate structures followed by selection of one, after which the discarded models stop contributing.

Data-driven and grey-box marine models span recurrent architectures [16], physics-informed networks [17] and gradient-boosted ensembles [18], and our Level 3 follows the neural identification literature [19] and its delayed-input variants [20]. The distinction relevant to control is between one-step prediction, which reads measured state history at every sample, and recursive simulation, which feeds predicted states back and is what a design loop requires. Reporting practice is uneven on this point, though Woo *et al.* [16] do validate through forward simulation. Three gaps follow: sufficiency is resolved by selecting one model rather than retaining the set, evaluation is inconsistent about the one-step versus recursive distinction, and the small rudder-steered vessel is underrepresented relative to differential-thrust research platforms.

III. Model Spectrum Methodology

Each model is evaluated by what it contributes to the PI-plus-feedforward loop of Sec. I rather than by goodness of fit: Level 1 settles the loop structure, Level 2 says where nonlinear feedforward is needed and Level 3 bounds what a black-box alternative would buy.

A. Level 1: First-Order Linear SISO Models

Level 1 treats surge and yaw as decoupled first-order channels,

$$G_v(s) = \frac{K_v}{\tau_v s + 1}, \qquad G_r(s) = \frac{K_r}{\tau_r s + 1}, \tag{1}$$

mapping normalized throttle to forward speed and normalized rudder to yaw rate. Both are identified as discrete autoregressive models with exogenous input, ARX(n_a, n_b) with $n_a = n_b = 1$, so that $y(k) = a_1 y(k-1) + b_1 u(k-1)$ with y the channel output and u its normalized command. The two coefficients are fitted by least squares on the 10 Hz record and converted to continuous time. The parameters map straight onto a compensator. τ is the open-loop pole the controller has to work against, so it sets the scale of closed-loop bandwidth worth asking for, and K sets the loop gain. A PI structure follows directly, and internal model control tuning fixes its gains in closed form, $k_p = \tau/(K\lambda)$ and $k_i = 1/(K\lambda)$ for a chosen closed-loop time constant λ. Sec. III poses Diagnostic 4 against exactly this compensator.

B. Level 2: Nonlinear State-Space Model

Level 2 retains the decoupled channel structure but replaces linear damping in surge with quadratic hydrodynamic drag,

$$\dot{v} = k_T\, u - d_2\, v|v|, \tag{2}$$

$$\dot{r} = k_\delta\, \delta - d_r\, r, \tag{3}$$

with u the normalized throttle and δ the normalized rudder. The yaw channel keeps a constant k_δ, which is a local approximation rather than a physical claim: rudder force grows with the square of flow speed, so k_δ must vary with forward speed, and the value identified here is the one holding near the speed at which the rudder steps were flown. Diagnostic 1 returns to this.

Identification is two-phase, and the phasing is what makes it work. Fitting (2) jointly for k_T and d_2 is poorly conditioned, because thrust and drag are both large and nearly cancelling during steady runs, so the data constrains their difference far better than either term. Instead d_2 is estimated from coasting segments alone, where $u = 0$ and (2) reduces to $\dot{v} = -d_2 v|v|$ with a single unknown. With d_2 fixed, k_T follows from the powered segments by linear least squares.

C. Level 3: Black-Box NARX Model

Level 3 is a fixed-tap residual NARX multilayer perceptron with 19 lagged inputs, one hidden layer of 64 tanh units and two outputs which are state increments rather than absolute states. State history extends 0.5 s into the past, throttle history 2 s and rudder history 1 s. The lag structure is a project-specific implementation choice consistent with delayed-input NARX models [20]. Being multi-input multi-output, the model is free to discover the surge-yaw coupling absent in Levels 1 and 2.

Level 3 is a compact black-box reference point in the comparison and no claim is made that this architecture is optimal. Its role as an exemplar is to bound the benefit of added complexity at the available data scale. Evaluation reports one-step prediction and recursive simulation separately, following the distinction drawn in Sec. II.

D. Evaluation Protocol

Two scores are reported throughout and they are not interchangeable. One-step prediction advances the model a single sample from measured state history, so every prediction begins from the truth. Recursive simulation initializes once from measured history, then feeds the model's own output back as its state while replaying the throttle and rudder actually recorded, and runs to the end of the segment without further correction. Level 3 initializes from 2 s of measured history, the span of its longest input tap; the structured levels need one sample. No controller is present in either case. The commands are those the operator or autopilot issued during the trial, replayed open loop, so recursive simulation asks whether the model can stand in for the vehicle, not how a control loop behaves.

Fit is reported per channel as $R^2 = 1 - \sum_k (y_k - \hat{y}_k)^2 / \sum_k (y_k - \bar{y})^2$ against the measured signal, so zero is the score of predicting the channel mean and negative values are worse than that. Levels 1 and 2 are scored in-sample on the trial they were identified from, and the Level-3 figures state which partition each uses, since the two are not comparable on equal terms.

E. Cross-Model Diagnostics

The value of the model spectrum is the comparison across levels rather than the three fits individually. Held-out prediction error is the usual check in machine learning, where the prediction is the deliverable. Here the coefficients are a deliverable too, and a maneuvering model can score well on held-out data while its individual coefficients remain physically meaningless [1]. A coefficient that cannot be interpreted cannot inform a control gain, so the four diagnostics below compare across model structures instead.

The four are not an arbitrary list. Levels 1 and 2 differ in exactly one term, the form of the surge damping, and they share exactly two simplifying assumptions, that the two channels are decoupled and that rudder authority does not vary with forward speed. One diagnostic tests the term that separates the levels, two test the assumptions both levels make, and the fourth asks whether any of it reaches the compensator. The first three therefore ask whether a term is present in the data; the fourth asks whether a term that is present changes the loop design. Each is reported with the control consequence that would follow from a positive result, since a diagnostic that cannot change a design decision is not worth running.

Diagnostic 1, speed-dependent rudder gain. Rudder force grows with the square of flow speed, so steering authority varies with forward speed, and Sec. I treats that coupling as what makes rudder-steered craft the interesting design case. The diagnostic asks whether the available data resolves the effect well enough to carry it into the model. Replace the constant k_δ in (3) with a speed-scaled $k_\delta v$ and test whether the yaw-rate fit improves. Gain scheduling is well within the capability of a production autopilot, so the question is not whether the controller can accommodate the effect but whether the model can say what schedule to use; a constant k_δ cannot, which leaves the schedule to be found by tuning at each speed on the water. The alternative tested is a partial one, since Nomoto scaling gives $k_\delta \propto v^2$ with $d_r \propto v$ [6] while the regression scales only the input gain and only linearly, so a null here is weaker evidence than a null from the fully scaled model.

Diagnostic 2, drag nonlinearity. The form of the surge damping is the only structural difference between Levels 1 and 2, so this diagnostic is the warrant for Level 2 existing at all. Were the damping linear, Level 2 would collapse onto Level 1 and the spectrum would hold two distinct models rather than three. Fit linear and quadratic damping to the coasting deceleration and compare. A dominant quadratic term calls for nonlinear feedforward on the speed loop.

Diagnostic 3, rudder-induced surge drag. This is the reverse coupling to Diagnostic 1, and together the two test the decoupling assumption in both directions: whether speed changes steering, and whether steering costs speed. Correlate δ^2 against the surge residual of the Level-2 fit. A correlation would make every rudder command a disturbance to the speed loop, which a cross-channel feedforward term could reject.

Diagnostic 4, is the added fidelity inside the robustness margin? The first three diagnostics ask whether a term is real. This one asks whether a real term changes the design, which is the case the sufficiency literature of Sec. I leaves untested. The robustness margin of a compensator is a budget for model error, so the test is whether the Level-1 to Level-2 discrepancy fits inside the margin of a controller designed from Level 1 alone. The construction has four steps.

Step 1. Design a nominal compensator from Level 1 only, in the target architecture of Sec. I, so that the test measures a change the deployed autopilot would experience rather than a property of some auxiliary controller. We use an internal model control PI, $C(s) = (\tau_v s + 1)/(K_v \lambda s)$, which cancels the Level-1 pole and gives a nominal loop $L = 1/(\lambda s)$ and complementary sensitivity $T(s) = 1/(\lambda s + 1)$, with λ the intended closed-loop time constant.

Step 2. Linearize Level 2 about operating points spanning the tested envelope. For surge, $\partial(v|v|)/\partial v = 2v_0$ for $v_0 > 0$, so (2) gives a speed-scheduled first-order plant

$$G_2(s, v_0) = \frac{K_2(v_0)}{\tau_2(v_0)s + 1}, \quad \tau_2 = \frac{1}{2d_2 v_0}, \quad K_2 = \frac{k_T}{2d_2 v_0}, \tag{4}$$

in which both the gain and the time constant fall as $1/v_0$.

Step 3. Form the multiplicative model error $\ell(\omega, v_0) = |G_2(j\omega, v_0)/G_1(j\omega) - 1|$ and test robust stability, $\|\ell T\|_\infty < 1$.

Step 4. Report the operating range over which the test passes. The output is not a yes or no but the speeds at which Level 1 stops being sufficient, which is an actionable number for a control designer and a direct answer to "enough for what."

Fig. 1. The experimental platform, a Pro Boat Blackjack 42 monohull of 1.07 m length, afloat during a field trial. Thrust comes from a single propeller and steering from a single rudder, both at the stern. The autopilot, the GNSS antenna and the radio links are mounted above the deck on the aft half of the hull.

Step 3 alone turns out to be insufficient, and Sec. V reports why. Robust stability is a statement about whether the loop remains stable, and quadratic drag is stabilizing, so at high speed the Level-1 design becomes sluggish rather than unstable and the stability test keeps passing. We therefore add a performance criterion alongside it: apply the Level-1 compensator to $G_2(s, v_0)$ and compare the achieved closed-loop bandwidth against the design intent $1/\lambda$. The pair of criteria bounds sufficiency from both sides.

IV. Experimental Platform and Data Collection

A. Vehicle

The platform, shown in Fig. 1, is a Pro Boat Blackjack 42, a production radio-controlled monohull of 1.07 m (42 in) length, instrumented to a mass of approximately 5 kg. Propulsion is single-screw: one four-pole water-cooled brushless motor turning a single propeller through a 160 A water-cooled electronic speed controller, with a single rudder driven by one digital servo and an 8S lithium-polymer pack supplying power. The vehicle is therefore underactuated in exactly the sense of Sec. I, with forward thrust on one channel and steering moment on another whose authority depends on forward speed.

Choosing a production hull rather than a purpose-built platform is deliberate. Vessels of this class are what the small-USV installed base looks like, and the identification problem is posed by the vehicle that exists rather than by one designed to be identifiable.

B. Autopilot, Sensing and Logging

Guidance, control and logging run on a Cube Orange autopilot executing ArduRover 4.6.3 (commit `3fc7011a`), with Mission Planner as the ground station. Position and ground speed come from a HERE3 GNSS receiver over DroneCAN, and angular rates from the autopilot's internal inertial measurement unit.

The actuator configuration is confirmed by the parameter set: `SERVO1` is assigned to ground steering and `SERVO3` to throttle, and every other servo output is disabled. There is one thrust input and one steering input, with no differential-thrust path. Commands are logged as `RCOU_C3` (throttle) and `RCOU_C1` (rudder) at 1100–1900 μs PWM and normalized to $[-1, 1]$ about the 1500 μs neutral point. Measured outputs are `GPS_Spd` ground speed at 5 Hz and `IMU_GyrZ` yaw rate at 137 Hz. The trials used here are flown in manual mode, so the logged commands are open-loop operator inputs rather than the output of any autopilot loop.

The autopilot's stabilization layer is the design target named in Sec. I, and it is worth stating concretely because it is what the model levels are ultimately for. Two independent loops run at that layer: throttle to forward speed (`ATC_SPEED_*`) and rudder to yaw rate (`ATC_STR_RAT_*`), each nominally a PID with an added feedforward term. On this vehicle the derivative gain of both loops is set to zero, so the deployed compensator is proportional-integral plus feedforward, matching the structure assumed in Sec. III. The remaining gains are $P = 0.4$, $I = 0.1$, $\mathrm{FF} = 0.2$ on the speed loop and $P = 1.0$, $I = 0.77$, $\mathrm{FF} = 0.2$ on the yaw-rate loop.

The configured cruise speed for the vehicle is 5 m/s. The identification trial described below spans 0–2.9 m/s, so it covers roughly the lower half of the speed range the vehicle is intended to operate over. This bears on Diagnostic 4 in Sec. V-D, whose sufficiency interval is narrower still.

C. Data Cleaning

Multi-rate streams are aligned to a common 10 Hz grid by linear interpolation. Records are segmented at mode changes, signal-freshness discontinuities and quality flags, so that no segment spans a transition that the models are not intended to represent.

D. Datasets

Levels 1 and 2 are identified from a single step-response trial, a 130 s working window drawn from a 341 s recording, covering throttle 0–39% and speed 0–2.9 m/s and comprising 1,300 samples at 10 Hz. One 29.3 s gap caused by vessel repositioning is handled by segmentation. This is intentionally minimal, a proof of concept for the workflow rather than a characterization of the vehicle.

Level 3 draws on a larger corpus: 44,133 samples in 70 slow-region segments from 65 source logs. The slow-region rule excludes a complete segment once speed stays above 3.4028 m/s for at least 1 s continuously. Allocation is grouped by source log, so every segment from one recording falls in a single partition and no recording is split across training and validation. Training uses 37 logs, being 40 segments and 28,023 rows. The remaining 28 logs, 30 segments and 16,110 rows are pooled for the evaluation reported below, and that pool needs describing accurately rather than as a held-out set. It combines three partitions with different histories: 8 logs used for early stopping and hyperparameter selection, 9 used to choose among candidate model families, and 11 nominally reserved as a closed test set that was in fact opened before the paper checkpoint was fixed. The checkpoint itself was selected manually after that broader evaluation rather than by the pre-declared tuning rule. Metrics on this pool are therefore optimistic and we treat them as descriptive rather than as an unbiased estimate of generalization. Ten complete high-speed segments consistent with planing are held aside for later regime-spanning work and are not used in the results below. That label is empirical GPS speed over ground rather than a verified physical planing transition.

A further 17 records, 49 minutes and 29,524 samples at 10 Hz, support one check reported in Sec. V-D. They reach 6.0 m/s and include reverse throttle, so they cover the upper half of the configured operating range that the identification trial never enters. Seven of them already appear in the Level-3 corpus above; the remainder are excluded from it by the slow-region rule, which is precisely why they carry the speed variation the check requires. Much of its rudder activity occurs in the autopilot's rate-controlled mode, where the logged rudder is a controller output and the firmware applies its own speed scaling. Those records are excluded from what follows, since identifying a speed-dependent rudder gain from them would partly recover the controller rather than the vehicle.

For direct comparison across levels, the frozen Level 3 model is also evaluated on the same 130 s trial used for Levels 1 and 2. That trial is not part of the Level-3 corpus, which we confirmed by cross-correlating it against every corpus segment, so it is genuinely unseen by Level 3 while Levels 1 and 2 are identified from it. Its 20-sample lag history leaves 1,279 scored rows against the 1,300 rows available to Levels 1 and 2.

V. Results

A. Level 1

Ordinary least squares on the step-response trial gives

$$G_v(s) = \frac{6.42}{1.95s + 1}, \qquad G_r(s) = \frac{0.771}{0.471s + 1}, \tag{5}$$

with simulation $R^2 = 0.930$ in surge and 0.826 in yaw rate. These are in-sample figures. The single trial is short enough that withholding part of it would leave too little excitation to identify against, so the fit and the score both use all 1,300 samples, and the same holds for Level 2. They measure how well the structure can represent this trial, not how well it generalizes. The surge time constant of 1.95 s and the yaw time constant of 0.471 s differ by a factor of four, which alone settles the loop structure: the two channels can be given separate loops at separate bandwidths.

B. Level 2

Two-phase identification gives

$$\dot{v} = 3.963\,u - 0.438\,v|v|, \tag{6}$$

$$\dot{r} = 1.474\,\delta - 1.912\,r, \tag{7}$$

with $R^2 = 0.909$ in surge and 0.820 in yaw rate. Both are marginally below Level 1 on this trial, which is expected: Level 1 is fit to minimize exactly this error on exactly this data, while Level 2 estimates its drag coefficient from coasting segments alone. The useful comparison is what the structure exposes rather than the R^2, and Sec. V-D takes that up.

Estimating d_2 from coasting is what makes the fit usable. The joint fit is poorly conditioned for the reason given in Sec. III, and the resulting parameters are sensitive to the powered-segment weighting in a way the coasting estimate is not.

C. Level 3

On the full slow-region validation set of 30 segments from 28 held-out logs, the one-step macro-NRMSE is 0.084 and the recursive macro-NRMSE is 0.425. Speed R^2 falls from 0.998 one-step to 0.836 recursive, and yaw-rate R^2 from 0.991 to 0.768. Recursive simulation covers every validation sample. Fig. 5 shows the surge channel across that set.

On the common trial Level 3 reaches one-step R^2 of 0.998 in speed and 0.989 in yaw rate. Under recursive simulation on that same trial it falls to -1.988 in speed and 0.259 in yaw rate, a negative R^2 meaning the model predicts worse than the mean of the measured signal. Fig. 6 shows the surge trajectory drifting below zero and staying there for tens of seconds.

The gap between one-step and recursive prediction is measured within a single model on identical data, so it is unaffected by how the models were fitted, and it is large. But the mechanism is specific enough to name, and naming it changes what the comparison means.

A long recursive rollout is governed by the model's equilibrium: hold the throttle constant and whatever speed the model settles to is what the rollout reports. Driving each level from rest at constant throttle for 60 s gives Fig. 3. Levels 1 and 2 have an equilibrium by construction, $v = K_v u$ and $v = \sqrt{k_T u / d_2}$ respectively, and both track the measured steady speeds, with a root-mean-square error against measurement of 0.33 m/s for Level 1 and 0.18 m/s for Level 2. The NARX does not. Over the throttle band where the trial spends most of its time it settles *below zero*, returning -0.07 m/s at a throttle of 0.20 where the vehicle measurably does 1.22 m/s, and its steady-state map is not monotone in throttle, which no physical vessel can be. Its error against measurement is 0.98 m/s.

The cause is the training data rather than the architecture. The corpus is 16.4% powered and holds the throttle steady for a 90th-percentile 1.6 s against a vehicle time constant of two to four seconds, so it almost never reaches equilibrium; the comparison trial is 69.8% powered and holds throttle for ten seconds at a time. The model was fitted almost entirely to transients and was never shown what steady state looks like, so it had no basis on which to learn one. Two retrains confirm that training alone does not repair this: extending the rollout horizons from 5 s to 40 s, and refitting on powered-only sub-segments, both leave the steady-state map non-monotone and neither improves the trial.

This bounds what Fig. 6 can be said to show. It is not a comparison of structured against black-box modeling, because it places two models that possess an equilibrium against one that does not. We make no claim from it about model class.

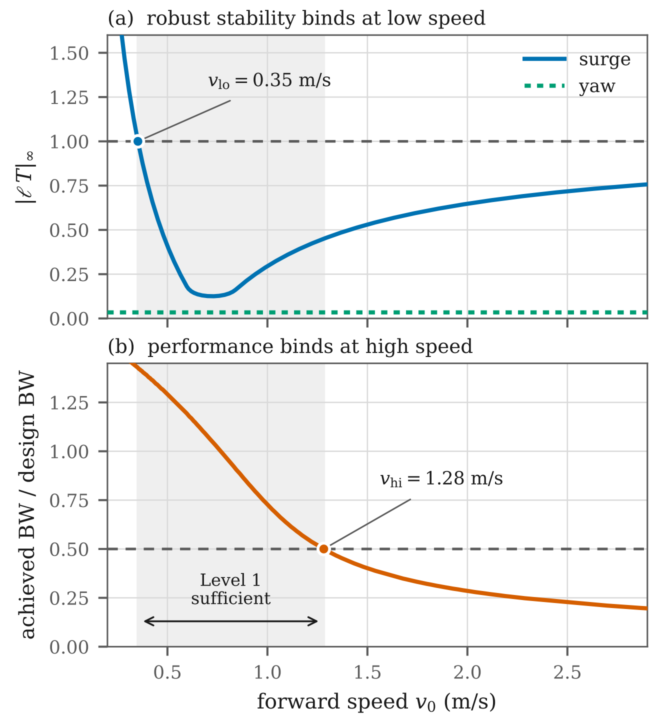

Fig. 2. Diagnostic 4. (a) Robust stability of the Level-1 compensator against the Level-2 linearization binds at low speed, where the quadratic-drag linearization is singular. (b) Achieved closed-loop bandwidth binds at high speed. The shaded band is the interval over which the Level-1 model supports the design. The yaw channel (a, dotted) never approaches the limit, so Level 2 changes nothing for yaw control.

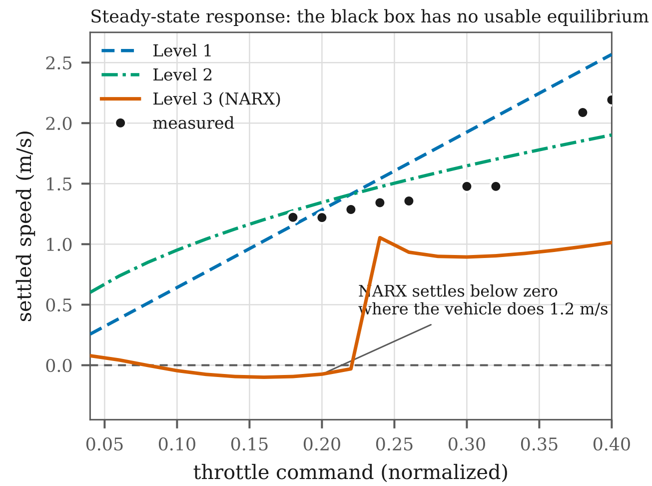

Fig. 3. Steady-state response of each level, obtained by driving the model from rest at constant throttle for 60 s and recording where it settles, against speeds measured on the trial. Levels 1 and 2 have an equilibrium by construction and track the measurements. The NARX settles below zero across the band the trial occupies and is not monotone in throttle.

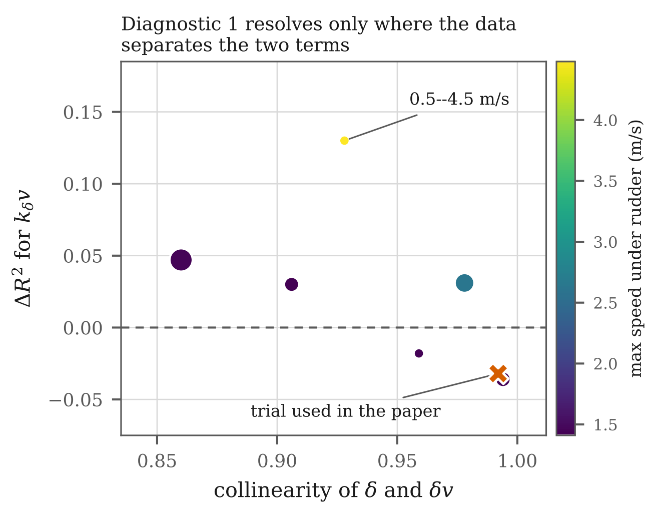

Fig. 4. Diagnostic 1 repeated on each manual-mode record of the supplementary corpus. The horizontal axis is the correlation between δ and δv over the samples with rudder applied, so records to the right cannot separate a constant gain from a speed-scaled one. Marker area is record length and colour is the highest speed reached under rudder. The cross marks the identification trial used throughout the paper.

What it does show is narrower and, for a practitioner, more useful: a model can be the best one-step predictor available and still be unusable in the closed loop, and reporting one-step accuracy alone will not reveal that.

D. Cross-Model Diagnostic Summary

Diagnostic 1 finds no improvement from a speed-dependent rudder gain ($\Delta R^2 = -0.03$). The trial holds forward speed nearly constant during the rudder steps, so δv and δ are very nearly the same regressor and no fit can separate them. The partial scaling noted in Sec. III is a second candidate explanation. Either way the result flags a gap in data coverage rather than absence of the effect, and it is the primary target of the next collection campaign, whose rudder sweeps at fixed speeds should test the full scaling rather than the input gain alone.

The supplementary corpus lets that attribution be tested rather than asserted. Measuring the collinearity of δ and δv on each manual-mode record and repeating the diagnostic gives Fig. 4. The identification trial sits at a correlation of 0.99, among the most collinear records available, and the records that separate the two terms are the ones in which the speed-scaled gain improves the fit. The clearest case holds rudder from 0.46 to 4.48 m/s and returns $\Delta R^2 = +0.13$. We do not claim the effect is thereby established: averaged over the manual-mode records the improvement is only $+0.02$, a single record carries most of it, and each fit is in-sample. What the comparison does establish is that the diagnostic's power is governed by how far speed varies while the rudder is deflected. That converts the explanation offered above from an assertion into a measurement, and it specifies what the rudder sweeps have to achieve: not more data, but rudder applied across separated speeds.

Diagnostic 2 finds quadratic damping clearly dominant, with $\Delta R^2 = +0.20$ over the linear fit, $d_1 = -0.015$ and

$d_2 = 0.448$ m^{-1} on the coasting segments. The two-phase identification of Sec. III returns $d_2 = 0.438$ m^{-1} from the same segments under the Level-2 parameterization, and the agreement between the two is a weak consistency check rather than an independent one. This is the direct empirical warrant for the Level-2 structure.

Diagnostic 3 finds rudder-induced surge drag negligible, with a correlation of $r = -0.08$ between δ^2 and the surge residual. The coupling term is not warranted by this data.

Diagnostic 4 is reported in Fig. 2, computed with $\lambda = 1.0$ s. The result is an interval rather than a threshold. Robust stability fails below $v_0 = 0.35$ m/s, and this lower bound has the closed form

$$v_{\text{lo}} = \frac{k_T}{4 d_2 K_v}, \tag{8}$$

which is independent of λ because $|T| \le 1$ places the supremum of $\ell|T|$ at DC. The mechanism is the $1/v_0$ singularity in (4): quadratic drag provides no linear damping at zero speed, so the linearized gain and time constant both diverge and no first-order fit can track them.

At the upper end the stability test never fails. At 2.9 m/s the model error is still within budget at $\|\ell T\|_\infty = 0.76$. What fails is performance: the Level-1 compensator applied to the Level-2 plant retains only 20% of its intended bandwidth at 2.9 m/s, crossing 50% at $v_{\text{hi}} = 1.28$ m/s. Quadratic drag is stabilizing, so the Level-1 design at speed is conservative rather than unsafe. The failure mode is sluggishness.

Taken together, the Level-1 surge model supports the control design only over roughly $[0.35, 1.28]$ m/s, under half of the tested envelope. The two bounds are not equally firm. The lower one follows from $\|\ell T\|_\infty = 1$ and has the closed form above, so it is a property of the two models. The upper one depends on choosing 50% of design bandwidth as the point where degradation becomes unacceptable, which is a convention rather than a result: a 70% criterion moves it to 1.03 m/s and a 30% criterion to 1.91 m/s. What the diagnostic supplies is the curve in Fig. 2(b) and the observation that degradation is well under way inside the tested envelope; the single number requires the designer to name a tolerance. Both bounds are insensitive to the design knob, since sweeping λ from 0.5 to 3 s moves v_{hi} only between 1.23 and 1.61 m/s and leaves v_{lo} unchanged.

For yaw the picture is different. Level 2 linearizes to $r/\delta = 0.771/(0.523s + 1)$ against Level 1's $0.771/(0.471s + 1)$: identical DC gain and an 11% difference in time constant, giving $\|\ell T\|_\infty = 0.034$. Level 2 buys nothing for yaw control across the whole envelope, and we say so rather than let the added structure imply otherwise.

VI. Discussion

A. What the Spectrum Reveals

For this vehicle class and at the speeds tested, surge and yaw are weakly coupled, so decoupled SISO controllers are a defensible starting point. Quadratic drag is significant, which means a speed-hold controller needs nonlinear feedforward and a purely linear controller will show speed-dependent steady-state error. Whether the rudder gain scales with forward speed can be neither confirmed nor denied from this data, and that is the single largest gap the expanded dataset must fill.

No one of the three models exposes all of this. Level 1 alone gives no reason to suspect the drag is quadratic. Level 2 alone does not reveal that its extra structure is inert for yaw. Level 3 alone would report an excellent one-step score and mislead anyone who took it to simulation.

B. Implications for Compensator Architecture

Level 1 supports a baseline PI structure with directly identified gains, valid over the interval Diagnostic 4 returns. Level 2 justifies nonlinear feedforward on the surge channel, and Diagnostic 4 says at what speed it becomes necessary; on the yaw channel it changes nothing, so the added structure should not be carried into the design. Gain scheduling on the rudder loop stays open, gated on Diagnostic 1. Level 3 bounds what lies beyond the fixed architecture: a black-box model free to discover the coupling the structured levels assume away is the natural precursor to model-predictive or learning-based control, and its recursive failure says plainly that the data does not yet support moving past PI plus feedforward. None of the three, though, delivers `ATC_SPEED` and `ATC_STR_RAT` gains that work on the water without adjustment. What the spectrum buys is the loop structure, a simulation to verify it in, a basis for comparing variants and a tuning session that can be planned rather than improvised.

C. Predictive Fit Against Simulation Usefulness

The Level-3 result is the paper's sharpest empirical claim. A model can be the most accurate one-step predictor in the comparison and simultaneously the least useful simulator, and the two properties are measured by numbers that differ by more than two units of R^2 on identical data. Any evaluation that reports only one-step accuracy is silent about the property a control designer needs, which is whether the model can stand in for the vehicle in a closed loop.

It is tempting to read this as showing that physical structure beats data volume, since Level 3 saw 28,023 training samples against the 1,300 used for Levels 1 and 2 and still simulated worst. That reading is wrong twice over. The levels were fitted and scored under different protocols, so the comparison cannot separate structure from sampling; and refitting Levels 1 and 2 on Level 3's own training partition and scoring all three on the same held-out segments reverses the ordering, giving recursive surge R^2 of $+0.53$, $+0.64$ and $+0.84$ respectively. On equal terms the network is the better predictor.

The defensible reading is about excitation rather than model class. What the network lacks is an equilibrium, and it lacks one because the records it learned from hold throttle steady for less than a time constant. A structured model does not need the experiment to contain steady running, because setting $\dot{v} = 0$ in (2) returns $v = \sqrt{k_T u / d_2}$ whether or not any trial ever sat there. That is the concrete advantage of writing the model

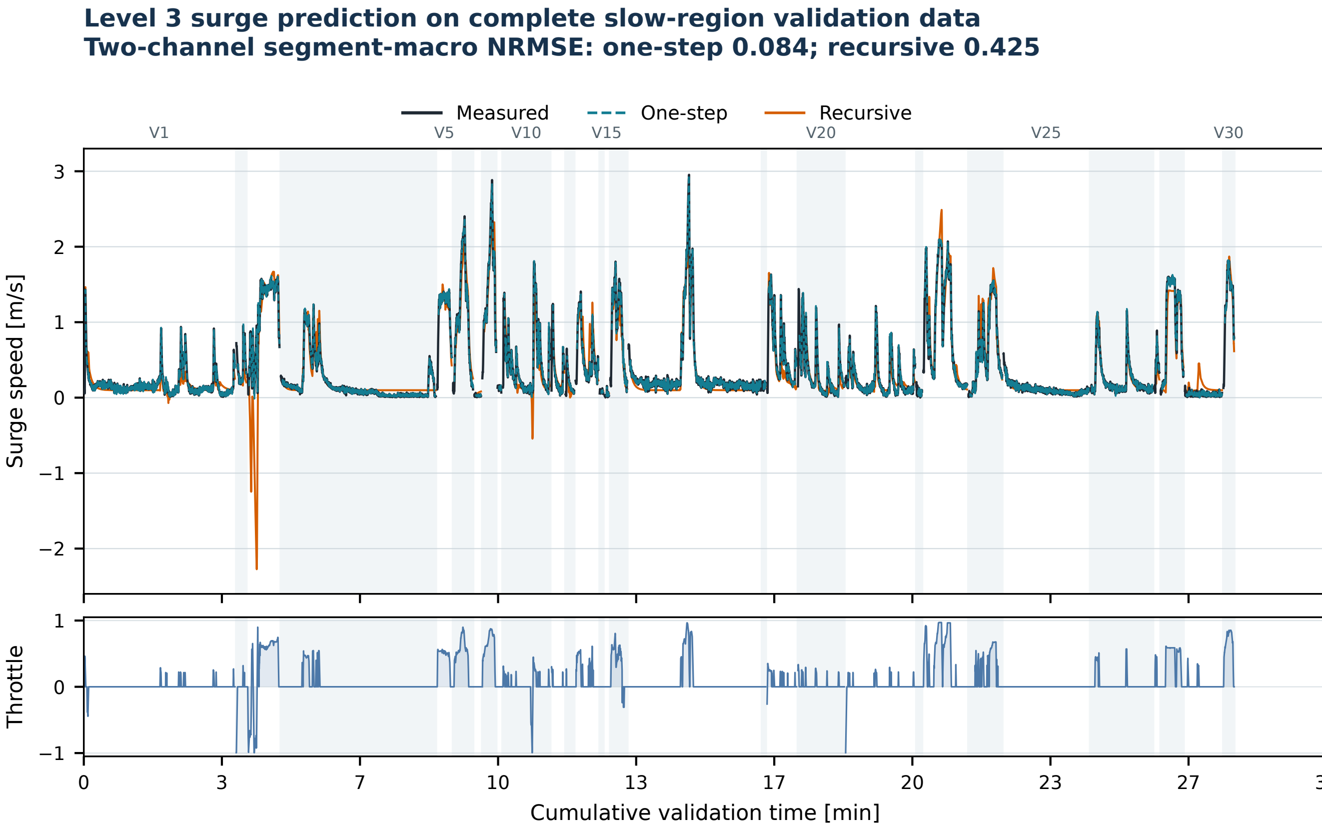

Fig. 5. Level 3 surge prediction on the complete slow-region validation set of 30 segments from 28 held-out source logs. Each segment is initialized with 2 s of measured history. One-step prediction continues to use measured state history; recursive simulation feeds predicted states back while replaying recorded throttle. Alternating backgrounds and short gaps mark segment boundaries. Reported NRMSE values are segment-macro averages over speed and yaw-rate errors normalized by training-only channel scales.

as an equation of motion, and it is a statement about how the learning problem was posed rather than about neural networks.

The same theme runs through Diagnostic 1, where the trial could not separate δ from δv, and through Diagnostic 4, where sufficiency is a property of the operating point. In each case what the model can tell a designer is bounded by what the experiment made visible.

D. Limitations

Levels 1 and 2 rest on a single trial covering 0–2.9 m/s and throttle to 39%, so the identified parameters are not a characterization of the vehicle across its envelope. Diagnostic 4 inherits that limitation, since the Level-2 linearization it uses is only as good as the parameters behind it. The common-trial comparison in Sec. V-C rests on one 130 s record, which is enough to demonstrate the divergence but not to quantify how often it occurs. Level 3's training corpus is broader but restricted to the slow region by construction, so nothing here speaks to behavior near or above the planing transition. Since the configured cruise speed is 5 m/s, none of the three levels is characterized over the upper half of the vehicle's intended operating range. All results come from a single vehicle, so generalization across hull form and scale within this class is untested here.

VII. Conclusion

We have presented a three-level model spectrum for small underactuated USVs, identified from field trials and evaluated on common data, and posed throughout against a single concrete question: what does a practitioner need in order to design, simulate and tune the speed and yaw-rate loops of a production autopilot? The central finding is that the spectrum collectively answers that question better than any single model does. Simple models are a persistent tool for this work rather than a stepping stone to be discarded once a better fit is available.

The margin-based sufficiency test converts Skelton's qualitative criterion into a computed interval for this vehicle class. For the surge channel it returns $[0.35, 1.28]$ m/s, bounded below by robust stability and above by loss of closed-loop bandwidth, and it reports that the same test is satisfied everywhere on the yaw channel, where the added Level-2 structure is inert.

Future work runs in three directions. The first is data: multiple trials spanning the full throttle range and dedicated rudder sweeps at fixed speeds, which is what resolves the

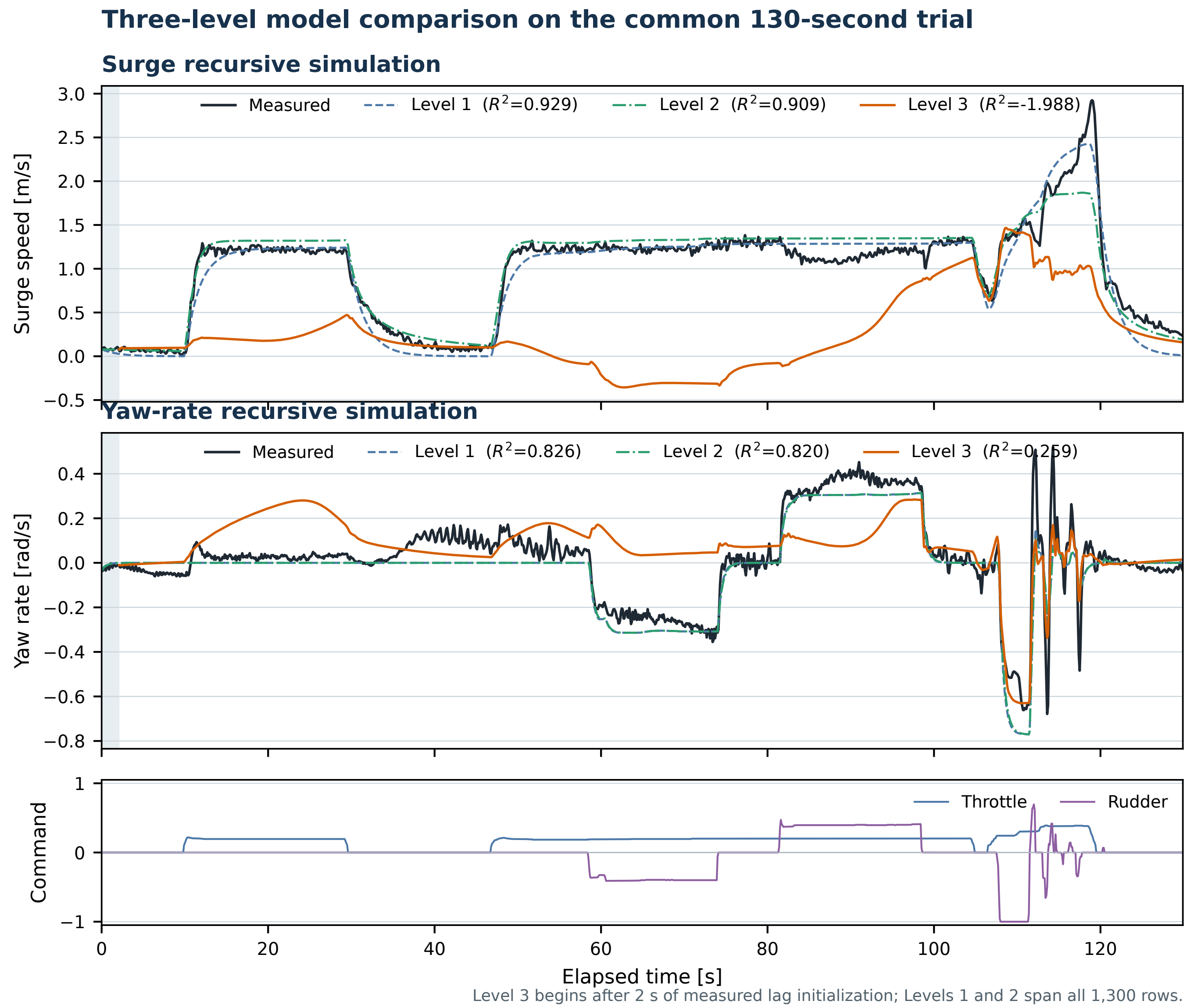

Fig. 6. All three levels under recursive simulation on the common 130 s trial. Levels 1 and 2 track the measured response; Level 3 diverges in surge, driving the predicted speed below zero for tens of seconds despite reaching the highest one-step accuracy of the three. Level 3 begins after 2 s of measured lag initialization, so it scores 1,279 rows against 1,300 for Levels 1 and 2.

speed-dependent rudder gain that Diagnostic 1 could not settle. The second is a stronger basis for cross-level comparison, since the present common trial is a single 130 s record and a multi-trial evaluation set would let the Level-3 divergence be quantified rather than demonstrated. The sensitivity of each level to training-set size belongs here too, particularly across displacement, semi-planing and planing regimes. The third is to treat the robustness margin as a formal sufficiency criterion across the spectrum, computing for each design decision the level at which further fidelity is provably inert, which would replace the expert judgment of Sec. II with a computable bound on how much model a given autopilot loop requires.

The dataset and analysis notebooks will be released.

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